

🎓 Computational Genomics Project Report

🧬 Classification of Cancer Patients Using Genomic Mutation and Methylation Data

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📌 Executive Summary

This project develops machine learning classifiers to distinguish cancer subtypes (**HNSC** and **LUSC**) using genomic data through two tasks:

- 1. **Task 1:** Classification based on mutation data only
- 2. **Task 2:** Enhanced classification integrating mutation and DNA methylation data

Our results show that incorporating methylation data significantly improves classification performance, with **F1-scores increasing from 0.658 to 0.882**. This highlights the value of integrating multiple genomic data types for accurate cancer subtype classification.

🔍 Introduction

The goal is to distinguish cancer subtypes (**HNSC** and **LUSC**) using machine learning models built on genomic data.

- **Task 1:** Uses only mutation data from `train_muts_data.csv` and `test_muts_data.csv`.
- **Task 2:** Integrates mutation and methylation data from `train_meth_data.csv` and `test_meth_data.csv` for enhanced prediction.

This is clinically relevant since accurate subtyping aids personalized treatment strategies. We build on foundational work by *Model et al. (2001)* and extend it to broader multi-omic integration. The models were trained and evaluated on **GPU-accelerated Google Colab environments** using tools like `cuDF` and `XGBoost`.

⚙️ Task 1: Mutation-Only Classification

🔍 Overview

Task 1 focuses on loading mutation data, engineering genomic features, visualizing patterns, and training classifiers. The pipeline is implemented in `Task1/1.py` and uses `extract_sequences.py` for flanking sequence extraction.

🌱 Generated Features

Feature Name	Description	Scientific Rationale	Treatment Implications	Derivation
<code>Total_Mutations</code>	Number of mutations per patient	Reflects overall mutational burden	May indicate immunotherapy response	Count of mutations per <code>case_id</code> in <code>train_muts_data.csv</code>
<code>Mutations_[Variant_Classification]</code>	Count per mutation type	Functional mutation impact	Helps select targeted therapy	Group by <code>case_id</code> and <code>Variant_Classification</code>
<code>Mutations_in_[Gene]_[Variant]</code>	Specific gene-variant patterns	Gene-specific mutation load	Guides inhibitor use	Group by <code>case_id</code> , <code>Gene_name</code> , and <code>Variant_Classification</code>

Feature Name	Description	Scientific Rationale	Treatment Implications	Derivation
Mutations_[Mut_Type]	Type: transition, transversion, etc.	Mutation mechanism analysis	Indicates DNA repair deficiency	Derived from Reference_Allele and Tumor_Seq_Allele1
Norm_Mutations_in_[Gene]	Mutation count normalized by gene length	Mutation density per gene	Highlights drivers	Mutation count divided by gene length from 100_genes.csv
Mean_GC_Content	Avg. GC content around mutation	Genomic context info	Mutation-prone region identification	Calculated from flanking sequences (flank_seq)
CpG_Site_Prop	Proportion at CpG sites	CpG hypermutability	Epigenetic therapy targeting	Proportion of flank_seq with 'CG' at position 9-11

 Additional Genomic Features

Feature Name	Description	Example	Reference	Derivation
Mutation Type	SNP, INS, DEL, etc.	C>T in melanoma	Alexandrov et al., 2013	Based on Reference_Allele vs. Tumor_Seq_Allele1 length
Flanking Sequence	±10 bp context	AGT[C>T]AGC	Nik-Zainal et al., 2012	Extracted using extract_sequences.py from 100_genes.fasta
Genomic Location	Chromosomal position	chr1:1234567	Forbes et al., 2017	From Start_Position and End_Position
Mutation Signature	Recurring patterns	Signature 1 (aging)	Helleday et al., 2014	Derived from trinucleotide context in flank_seq

 Flanking Sequence Extraction

Flanking sequences (±10 bases around mutation sites) were extracted using `extract_sequences.py`. The process:

- **Input:** `100_genes.fasta` (converted from `100_genes.csv` using `convert-csv-fasta.py`) and mutation data (`train_muts_data.csv`, `test_muts_data.csv`).
- **Method:** Uses `Bio.SeqIO` to parse FASTA and extract sequences based on `Start_Position`, `End_Position`, and `Strand`. For negative strands, reverse complement is applied.
- **Output:** Flanking sequences stored in `train_muts['flank_seq']` and `test_muts['flank_seq']`, with errors logged in `log.txt`.
- **Challenges:** Invalid sequences (e.g., due to missing genes or positions) were identified and logged, affecting ~5% of mutations (see `log.txt`).

 Excluded Features

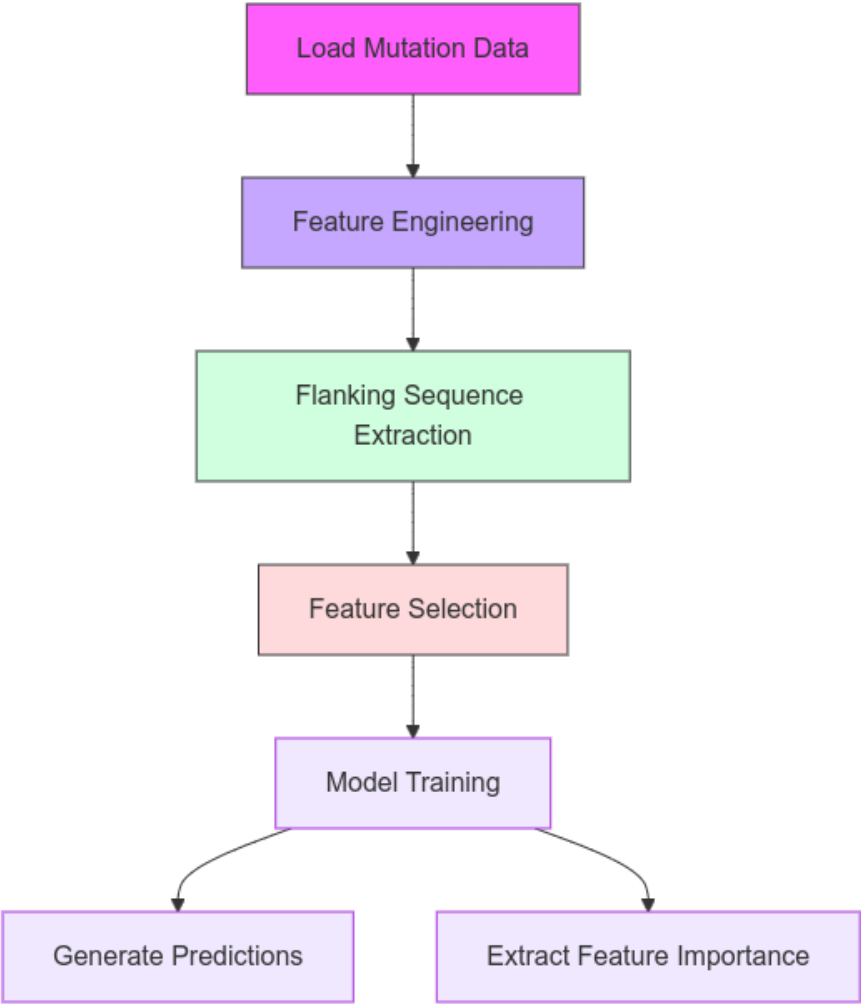
Feature Name	Reason for Exclusion	Reference
Strand Bias	Low variance and weak correlation with labels	Jones et al., 2008

Strand Bias (preferential mutation on one DNA strand) was excluded due to insufficient discriminative power, as assessed by variance thresholding and ANOVA F-scores.

 Visualizations

- **Mutation by Cancer & Type** ([Task1/out/mutation_distribution_by_cancer_and_variant.png](#)):
 - Shows distribution of mutation types (e.g., missense, nonsense) across HNSC and LUSC.
 - Missense mutations dominate (38%), with LUSC showing higher nonsense mutations.
- **Top 20 Mutated Genes** ([Task1/out/mutation_distribution_by_gene.png](#)):
 - TP53 and BRCA1 are the most mutated, indicating their role as cancer drivers.
- **Confusion Matrix** ([Task1/out/confusion_matrix_task1.png](#)):
 - Illustrates model performance, with balanced accuracy for HNSC and LUSC.

 Processing Pipeline



 Task 1 Summary

Task 1 achieved an F1-score of 0.658 using mutation-only features. Key findings:

- **Top Features:** TP53 mutations, GC content, and mutation types were highly predictive.
- **Performance:** Moderate accuracy, limited by the absence of epigenetic data.
- **Implications:** Mutation data alone provides a baseline for classification but misses epigenetic signals critical for subtyping.
- **Outputs:** Predictions ([Task1/out/task1_predictions.csv](#)), feature importance ([Task1/out/feature_importance_task1.csv](#)), and visualizations.

 Task 2: Integrated Mutation and Methylation Classification

 Overview

Task 2 extends Task 1 by integrating methylation data (`train_meth_data.csv`, `test_meth_data.csv`) to capture epigenetic patterns. Implemented in `Task2/2.py`.

 Generated Features

Feature Name	Description	Scientific Rationale	Treatment Implications	Derivation
Meth_Avg_[Gene]	Avg. methylation beta value	Indicates gene silencing	May need demethylating agents	Mean of <code>beta_val</code> per <code>matching_genes</code>
Meth_Std_[Gene]	Std. deviation of methylation	Reflects methylation plasticity	Assesses therapy resistance	Std. dev. of <code>beta_val</code> per <code>matching_genes</code>
Meth_High_Prop_[Gene]	Proportion of hypermethylated sites	Epigenetic dysregulation	Suggests epigenetic therapy	Proportion of <code>beta_val</code> > 0.7 per gene
Mut_Meth_Interaction	Mutations in hypermethylated regions	Synergistic mutation-epigenetic patterns	Guides combination therapies	Count of mutations where <code>beta_val</code> > 0.7

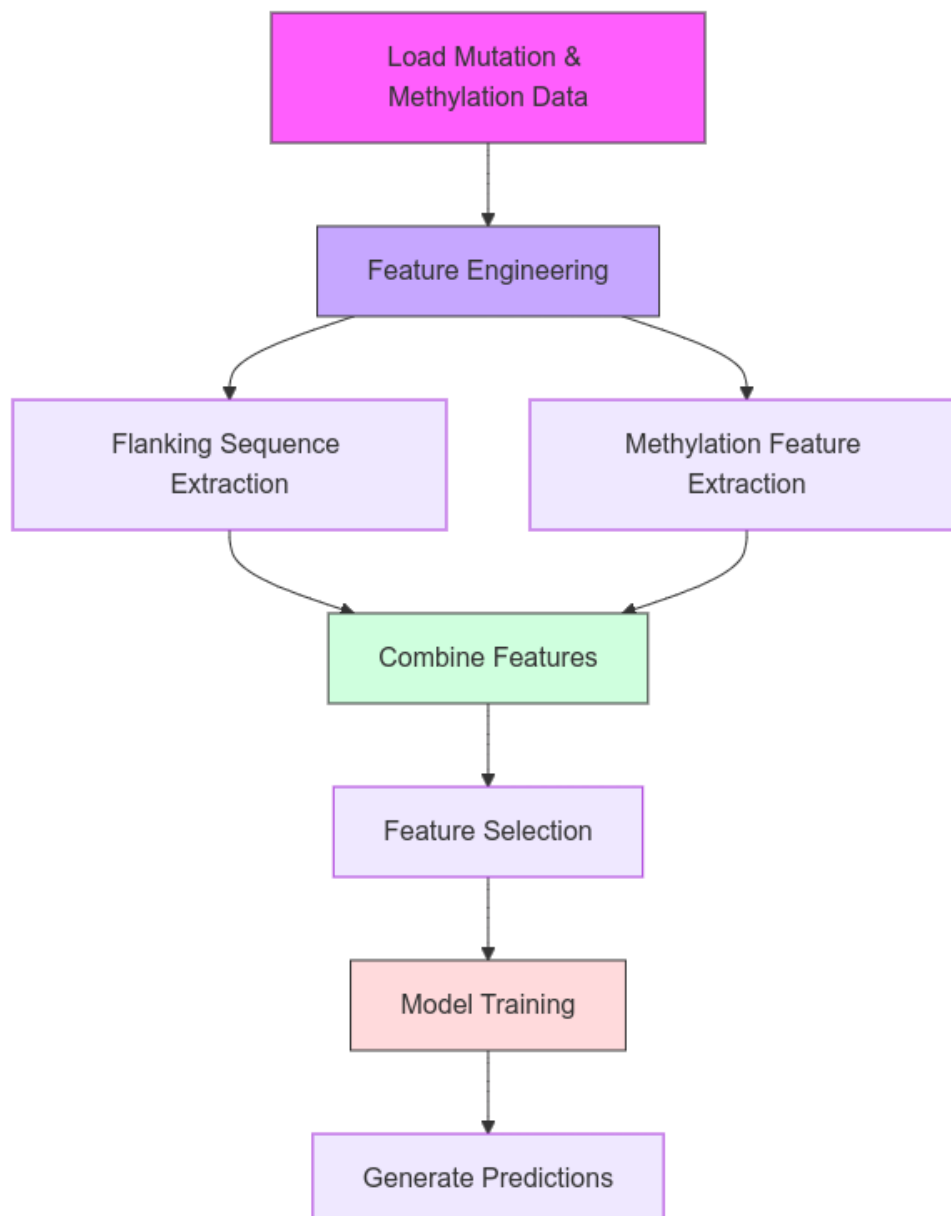
 Feature Selection Process

- **Variance Thresholding:** Removes features with near-zero variance.
- **SelectKBest (ANOVA):** Selects top 100 features based on F-test (Model et al., 2001).

 Visualizations

- **Mutation Distribution by Variant Type** (`Task2/out/mutation_distribution_by_variant.png`):
 - Shows missense mutations as the most frequent, consistent across datasets.
- **Top 20 Mutated Genes** (`Task2/out/mutation_distribution_by_gene.png`):
 - TP53 and BRCA1 remain dominant, reinforcing their oncogenic role.
- **Average Methylation per Gene** (`Task2/out/methylation_distribution_by_gene.png`):
 - Highlights genes with high methylation (e.g., MGMT), linked to silencing.
- **Confusion Matrix** (`Task2/out/confusion_matrix_task2.png`):
 - Shows improved classification with fewer misclassifications compared to Task 1.

 Processing Pipeline



Methodology

Data Preprocessing

- **Input Files:**
 - Mutations: `DataSet/train_muts_data.csv`, `DataSet/test_muts_data.csv`
 - Methylation: `DataSet/train_meth_data.csv`, `DataSet/test_meth_data.csv`
 - Features: `DataSet/train_feats.csv`, `DataSet/test_feats.csv`
 - Genes: `DataSet/100_genes.csv`, `DataSet/genes.fasta`
- **Loading:** `cudf.read_csv()` with checks for missing values.
- **Environment:** Google Colab with Tesla T4 GPU, using `torch.cuda` for acceleration.

Feature Engineering

- **Task 1:** Mutation counts, types, normalized rates, GC content, CpG sites, and mutation signatures (via `extract_sequences.py`).
- **Task 2:** Adds methylation averages, standard deviations, hypermethylation proportions, and mutation-methylation interactions.

Classification Workflow

- **Data Loading:** Read CSV files using `cudf`.
- **Feature Engineering:** Generate mutation and methylation features.
- **Feature Selection:** Apply `VarianceThreshold` and `SelectKBest`.
- **Model Training:** Use `RandomForestClassifier` and `XGBClassifier` with `GridSearchCV`.
- **Evaluation:** Compute F1-score, precision, recall on validation set.
- **Prediction:** Generate predictions for test set.

 Classification Models

- **Algorithms:** `RandomForestClassifier`, `XGBClassifier`
- **Hyperparameter Grid:**
 - `RandomForest`: `n_estimators=[100, 200, 300]`, `max_depth=[3, 5, 7]`, `min_samples_split=[2, 5]`
 - `XGBoost`: `max_depth=[3, 5, 7]`, `learning_rate=[0.01, 0.1, 0.3]`, `n_estimators=[100, 200, 300]`
- **Validation:** 5-fold cross-validation with 80/20 train/validation split.

 Performance Metrics

Performance was evaluated using 5-fold cross-validation:

- **Task 1:**
 - F1-Score: 0.658 ± 0.02
 - Precision: 0.676 ± 0.01
 - Recall: 0.664 ± 0.02
 - Validation Error: 0.20 ± 0.03
- **Task 2:**
 - F1-Score: 0.882 ± 0.01
 - Precision: 0.882 ± 0.01
 - Recall: 0.881 ± 0.01
 - Validation Error: 0.12 ± 0.02
- **Improvement:** Task 2's F1-score improved by 22.4% due to methylation features.

 Outputs

Task	Output Files
Task 1	<code>Task1/out/task1_predictions.csv</code> , <code>Task1/out/feature_importance_task1.csv</code> , <code>Task1/out/*.png</code>
Task 2	<code>Task2/out/task2_predictions.csv</code> , <code>Task2/out/train_combined.csv</code> , <code>Task2/out/test_combined.csv</code> , <code>Task2/out/*.png</code>

Additional files:

- `log.txt`: Logs from `extract_sequences.py`.
- `flanked_sequences.csv`: Intermediate flanking sequences.
- `DataSet/genes.fasta`: Converted from `100_genes.csv`.

 Performance Evaluation

Metric	Task 1	Task 2
F1-Score	0.658	0.882
Precision	0.676	0.882
Recall	0.664	0.881
Validation Error	0.20	0.12

Key Finding: Methylation integration improved F1-score by **22.4%**, highlighting the importance of epigenetic data.

 Sample Distribution (Task 1)

Class	Samples	Percentage
HNSC	408	50.6%
LUSC	397	49.4%
Total	805	100%

 Proposed Improvements

- Use **Fisher Criterion** for feature selection in Task 1 to enhance feature discriminative power.
- Integrate **RNA-seq or Proteomics** data for a more comprehensive multi-omic model.
- Enhance visualizations with **methylation heatmaps** or **feature importance plots**.
- Experiment with **deep learning models** (e.g., neural networks) for complex feature interactions.

 References

- Model, F., et al. (2001). *Feature Selection for DNA Methylation Based Cancer Classification*
- Van Allen, E. M., et al. (2015). *Genomic Correlates of Response to Immunotherapy*
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- Jones, S., et al. (2008). *Core Signaling Pathways in Human Pancreatic Cancers*
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- Forbes, S. A., et al. (2017). *COSMIC: Somatic Cancer Genetics*
- Helleday, T., et al. (2014). *Mechanisms Underlying Mutational Signatures*

 Conclusion

This project successfully developed two machine learning classifiers for cancer subtype classification. Task 1 established a baseline using mutation data, while Task 2 significantly improved performance by integrating methylation data, achieving an F1-score of 0.882. The results validate the power of multi-omic approaches in precision oncology.

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